TIAGO RODRIGUES DE ALMEIDA

Machine Learning & AI Specialist
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English and Portuguese

ABOUT ME

I have a Ph.D. in Computer Science, specializing in machine learning, and I have a strong foundation in deep learning. I am passionate about using AI to drive innovation and solve complex data-driven problems. I develop machine learning techniques to build scalable, high-impact, data-driven solutions.

github.com/tmralmeida
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scholar.google.com/citations?user=ORMNS9kAAAAJ&hl=en

WORK EXPERIENCE

Freelancing, Remote *Oct. 2025 - Present*
AI Consultant

- **Generative AI & Edge Computing:** Architected and deployed *SnapMade*, an end-to-end MVP that reconstructs 3D-printable meshes from single 2D images. Optimized large-scale Generative AI models (Python/PyTorch) for low-latency inference on ARM64 architecture (NVIDIA Jetson AGX Orin) and integrated a full-stack cloud backend using Supabase and Stripe.
- **Process Automation & Data Engineering:** Engineered a custom administrative automation platform using Python and Streamlit for a multi-national company. Developed complex logic to automate international labor allowances, salary calculations, and dynamic vacation tracking, transforming hours of manual spreadsheet work into a one-click PDF reporting system.
- **SME Strategic Consulting:** Partnering with small-to-medium enterprises to bridge the gap between academic research and production-ready products, specializing in the deployment of Large Vision Models (LVMs) and LLMs in resource-constrained environments.

Amazon, Barcelona, Spain *Oct. 2022 - April 2023*
Applied Scientist Intern

- Led the development of Transformer and ResNet-based models using quantile distributions to improve transit-time predictions (Python/PyTorch), resulting in €10M annual business impact.
- Engineered scalable real-time inference pipelines on AWS and developed Streamlit dashboards to visualize model behavior and uncertainty for finance and operations stakeholders.

University of Aveiro, Portugal *Sep. 2019 - Sep. 2020*
Research Fellow

- Architected a full-stack, low-cost localization and navigation system from scratch (C/C++), implementing deep learning models (Python/PyTorch) and trilateration techniques for self-localization.
- Developed deep learning-based object detection and decoding methods for static landmarks, conducting comparative analyses on edge devices (NVIDIA Jetson AGX Xavier) using DeepStream.

Renault, Aveiro, Portugal *July 2018 - Aug. 2018*
Electronics Engineer Intern (Computer Vision)

- Improved an artificial vision station for defect detection in a production line (Sherlock).

EDUCATION

Örebro University and Technical University of Munich (TUM) *Sep. 2020 - Sep. 2025*
Ph.D. in Computer Science (Machine Learning) – WASP Program and Research Stay

- Have shown that semantic attributes enhance trajectory prediction.
- Studied machine learning and deep learning techniques for time series analysis (Python/PyTorch).
- Wrote several scientific publications for international journals and conferences (e.g. RA-L).
- Teacher on the Computer Networks course.

University of Aveiro, Portugal *Sep. 2014 - July 2019*
M.Sc. in Mechanical Engineering – GPA: 16/20

- Masters in Robotics and Computer Vision applied to Autonomous Driving (19/20).
- Thesis: Developed a multi-camera and multi-algorithm ROS (C/C++) architecture for road perception onboard an autonomous vehicle.

SPECIALIZATIONS

Hugging Face: LLM and Agents Courses *July - Aug. 2025*

Wallenberg AI, Autonomous Systems and Software Program *Oct. 2020 - 2024*

- Deep Learning and GANs
- Learning Feature Representations
- Mathematics and Machine Learning
- Artificial Intelligence and Machine Learning
- Software Engineering and Cloud Computing

Amazon Web Services: AWS Technical Essentials *Nov. 2022*

CS50 from Harvard University: CS50x *June 2020*

NVIDIA: Deepstream for video analytics on Jetson Nano *March 2020*

Coursera *Dec. 2019 - Feb. 2020*

- Deep Learning Specialization
- Applied Machine Learning in Python Course
- Applied Plotting, Charting & Data Representation in Python
- Introduction to Data Science in Python
- TensorFlow in Practice Specialization

ROS Workshop: 25 hours learning Robot Operating System *Feb. 2019*

Wall Street English: First Certificate in English - B2 *Sep. 2019*

EXTRA

- Reviewer for CVPR, Nature Communications, ICCV, CoRL, ICRA, and RA-L *2020-Present*
- BE–Digital Lecturer *2025*
- Think Twice 40 hours programming (Python and C/C++) *2020*